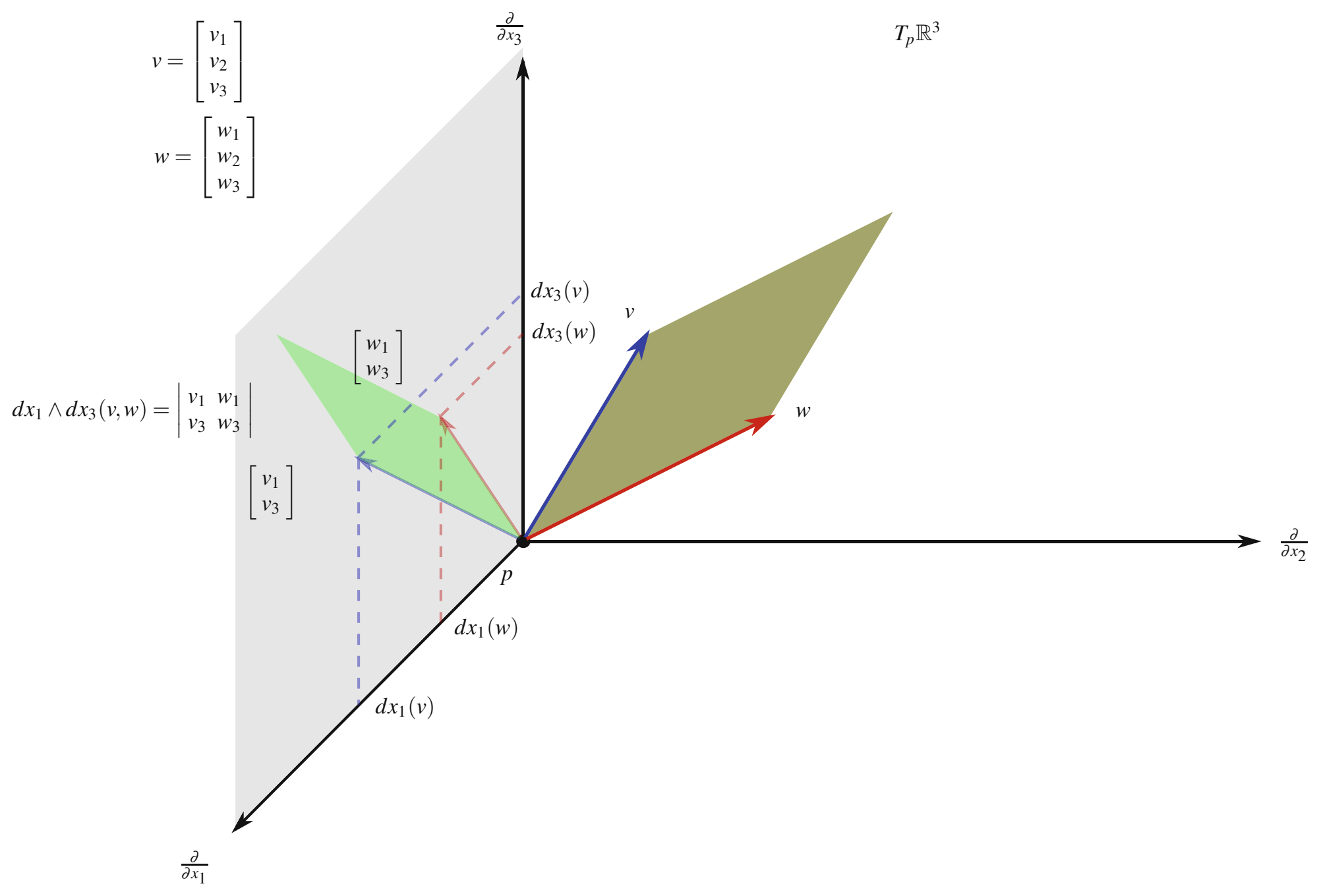

$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$
$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$
$dx_1 \wedge dx_3(v,w) = \begin{vmatrix} v_1 & w_1 \\ v_3 & w_3 \end{vmatrix}$
$\frac{\partial}{\partial x_3}$
$T_p\mathbb{R}^3$
$dx_3(v)$
$dx_3(w)$
v
w
$\begin{bmatrix} w_1 \\ w_3 \end{bmatrix}$
$\begin{bmatrix} v_1 \\ v_3 \end{bmatrix}$
$\frac{\partial}{\partial x_2}$
p
$dx_1(w)$
$dx_1(v)$
$\frac{\partial}{\partial x_1}$